A Krylov–Diameter Correspondence on Type II_∞ FRW Crossed Product Algebras*

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Abstract

We record a geometric–algebraic identity, the *Krylov–Diameter Correspondence*, on the type II_∞ crossed product $\mathcal{N} := \mathcal{A}(D_O)_{\mathrm{FRW}} \rtimes_\sigma \mathbb{R}$ of a doubly-bounded comoving causal diamond D_O in a conformally flat Friedmann–Robertson–Walker background, with $\mathcal{A}(D_O)_{\mathrm{FRW}}$ the local algebra of the conformally coupled massless scalar in the conformal vacuum (Theorem 3.5 of [7]). On a smooth seed inside the diamond, the Block A modular Krylov spread complexity C_k on $\mathcal{N}$, lifted from the Caputa–Magán–Patramanis–Tonni construction [2] via the Connes–Takesaki dual weight, satisfies in cosmic time t along the central worldline

$$\frac{1}{C_k(t)}\,\frac{\mathrm{d}C_k(t)}{\mathrm{d}t} \;=\; \frac{1}{R_{\mathrm{proper}}(\eta_c)} \;+\; O\big(C_k^{-1}\big),$$

where $R_{\mathrm{proper}}(\eta_c) = a(\eta_c)\,R_{\mathrm{conf}}$ is the proper diameter radius at the central Cauchy slice η_c. Both sides are spacetime scalars on the worldline. We tabulate four era/diamond combinations (radiation, matter, exact de Sitter; past-light-cone vs. Hubble-horizon diamonds) verified symbolically in sympy. The Hubble identity $H(t) = (1/C_k)\,\mathrm{d}C_k/\mathrm{d}t$ holds exactly in two of these and fails by an era-dependent $O(1)$ factor in the others; we do *not* claim a universal Hubble–Krylov identity. The Block A definition of C_k on a type II_∞ crossed product is novel beyond the type II_1 [1] and type III_1 [2] precedents and remains conjectural at the level of the universal operator-growth hypothesis on II_∞ KMS states.

1 Introduction

The crossed-product programme of Witten [8] and Chandrasekaran, Longo, Penington and Witten (CLPW) [4] produces, from a type III_1 local algebra $\mathcal{A}(\mathcal{O})$ and a one-parameter modular automorphism group σ, a type II von Neumann algebra of *gravitationally dressed* observables. The semifinite trace on the resulting type II factor is a quantitative entropy carrier and underlies recent advances in the generalised second law and entanglement entropy of subregions in quantum gravity.

In parallel, Caputa, Magán, Patramanis and Tonni (CMPT) [2] introduced *modular Krylov complexity* for the operator-spread under the modular flow on type III_1 algebras, and proved a universal late-modular-time slope $\lambda_L^{\mathrm{mod}} = 2\pi$ (their Eq. (4.16)/(5.4)). This is the modular analogue of the

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universal operator-growth hypothesis [6]. The slope 2π is intrinsic to the modular automorphism Δ^{is} of Tomita–Takesaki theory; it is independent of any background-spacetime time coordinate.

The present note connects these two strands in the simplest non-stationary cosmological setting where both make sense. Our starting point is the type II_∞ classification of the crossed-product algebra $\mathcal{N} := \mathcal{A}(D_O)_{\mathrm{FRW}} \rtimes_\sigma \mathbb{R}$ on a comoving causal diamond D_O in a conformally flat FRW background, established for the conformally coupled massless scalar in the conformal vacuum in [7, Thm. 3.5] via a conformal pullback to the Hislop–Longo modular flow on the Minkowski diamond [5]. On this II_∞ factor we consider the Krylov spread complexity C_k defined by Lanczos iteration on the lifted modular generator (Block A definition, Section 3); its rate of growth in cosmic time t along the central worldline is the object of interest.

The result we record is a clean geometric identity

$$\frac{1}{C_k(t)}\,\frac{\mathrm{d}C_k(t)}{\mathrm{d}t} \;=\; \frac{1}{R_{\mathrm{proper}}(\eta_c)} \;+\; O\big(C_k^{-1}\big), \tag{1}$$

where $R_{\mathrm{proper}}(\eta_c) = a(\eta_c)\,R_{\mathrm{conf}}$ is the proper diameter radius at the central Cauchy slice. This holds in any conformally flat FRW background covered by [7, Thms. 3.5–3.6, Cor. 3.7], *independent of the equation of state* and *independent of the particular convention chosen for the diamond*, provided $D_O \subset (\eta_i, \eta_f)$ with $a(\eta)$ smooth and strictly positive on $\overline{D_O}$ (i.e. bounded away from any FRW singularity).

The Hubble identity $(1/C_k)\,\mathrm{d}C_k/\mathrm{d}t = H(t)$ is an era- and diamond-convention-dependent *corollary* of (1), exact for radiation + past-light-cone and de Sitter + Hubble-horizon, and *wrong by a factor of* $1/2$ for matter + past-light-cone (Section 5). We make no universal Hubble–Krylov claim; the substantive content of the note is the geometric identity (1).

Plan. Section 2 recalls the type II_∞ classification of $\mathcal{N}$ from [7]; Section 3 states the Block A definition of C_k. The main theorem (Theorem 4) is in Section 4; Section 5 is the era-by-era table; Section 6 compares with [2]; and Section 7 lists open problems.

2 Setup: type II_∞ on the FRW comoving diamond

2.1 The local algebra and the conformal pullback

We adopt the framework of [7, §§2–3]. Let (M, g_{FRW}) be a spatially flat conformally flat Friedmann–Robertson–Walker spacetime with $\mathrm{d}s^2_{\mathrm{FRW}} = a(\eta)^2\big(-\mathrm{d}\eta^2 + \mathrm{d}\mathbf{x}^2\big)$ in conformal time $\eta \in I \subset \mathbb{R}$, with $a(\eta)$ smooth and strictly positive on the open interval I. We work with the conformally coupled massless scalar field φ ($\xi = 1/6$) in the conformal (Bunch–Davies) vacuum Ω_{FRW}, and with a doubly-bounded causal diamond D_O for a comoving observer at $\mathbf{x} = \mathbf{0}$, of conformal-time interval $[\eta_i, \eta_f] \subset I$ and central Cauchy slice $\eta_c := (\eta_i + \eta_f)/2$. The local von Neumann algebra $\mathcal{A}(D_O)_{\mathrm{FRW}}$ is generated, in the GNS representation $\pi_{\Omega_{\mathrm{FRW}}}$, by the smeared fields $\varphi(f)$ for $f \in C_c^\infty(D_O)$.

Theorem 1 ([7, Thm. 3.5] and [7, Thm. 3.6, Cor. 3.7])**.** *For any of the three FRW backgrounds $a(\eta) = a_0\eta$ (radiation, $\eta \in (0,\infty)$), $a(\eta) = a_0\eta^2$ (matter, $\eta \in (0,\infty)$), $a(\eta) = -1/(H_{\mathrm{dS}}\,\eta)$ (de Sitter Poincaré patch, $\eta \in (-\infty, 0)$), and any doubly-bounded comoving diamond D_O as above, the modular automorphism group σ^{FRW} of the pair $(\mathcal{A}(D_O)_{\mathrm{FRW}}, \Omega_{\mathrm{FRW}})$ is unitarily intertwined with the Hislop–Longo Möbius flow σ^{HL} on the conformally identified Minkowski diamond, has Connes spectrum $\Gamma(\sigma^{\mathrm{FRW}}) = \mathbb{R}$, and the crossed product*

$$\mathcal{N} \;:=\; \mathcal{A}(D_O)_{\mathrm{FRW}} \rtimes_{\sigma^{\mathrm{FRW}}} \mathbb{R}$$

is a type II_∞ factor.

2.2 The semifinite trace and standard form

By Connes–Takesaki, $\mathcal{N}$ carries a (semifinite, faithful, normal) canonical trace $\mathrm{tr}_{\mathcal{N}} := \phi_{\Omega_{\mathrm{FRW}}} \circ E$ obtained from the dual operator-valued weight to the modular flow [4, 8]. Let $L^2(\mathcal{N}, \mathrm{tr}_{\mathcal{N}})$ be the standard form Hilbert space of $\mathcal{N}$, with cyclic separating vector Ξ_ω representing the trace. We write $\Delta_{\mathcal{N}}$ for the modular operator of $(\mathcal{N}, \Xi_\omega)$; under the Connes–Takesaki dual-weight construction the modular generator decomposes as $K_{\mathcal{N}} = -\log \Delta_{\mathcal{N}} = K_{\mathrm{FRW}} \otimes \mathbf{1} + \mathbf{1} \otimes p_t$, where $K_{\mathrm{FRW}} = -\log \Delta_{\mathrm{FRW}}$ is the lifted modular Hamiltonian of the original type III_1 algebra $\mathcal{A}(D_O)_{\mathrm{FRW}}$ and p_t is the dual translation generator from the $\mathbb{R}$-factor in the crossed product. (We follow the sign conventions of [4, 8].)

2.3 The proper diameter radius

The diamond D_O has, on each constant-η Cauchy slice, an *intrinsic comoving radius* R_{conf} and an *intrinsic proper radius*

$$R_{\mathrm{proper}}(\eta) \;:=\; a(\eta)\, R_{\mathrm{conf}}, \tag{2}$$

which is a spacetime scalar (a function on $\overline{D_O}$, gauge-invariant under reparametrisations of η). At the central Cauchy slice $\eta = \eta_c$ we write $R_{\mathrm{proper}}(\eta_c)$. The prescription for R_{conf} in terms of η_i, η_f depends on the diamond convention; we discuss two natural choices in Section 5.

3 Block A definition of C_k on a type II_∞ crossed product

3.1 Definition

Following [2] for the type III_1 case, and lifting via the Connes–Takesaki dual weight to the type II_∞ standard form of $\mathcal{N}$, we define:

Definition 2 (Block A modular Krylov spread complexity on a type II_∞ crossed product)**.** Fix a smooth seed $|\mathrm{seed}\rangle := \pi_{\Omega_{\mathrm{FRW}}}\big(\varphi(f_0)\big)\, \Xi_\omega$ with $f_0 \in C_c^\infty(D_O)$, viewed as a vector in $L^2(\mathcal{N}, \mathrm{tr}_{\mathcal{N}})$. Run the Lanczos algorithm on $-iK_{\mathcal{N}}$: this generates an orthonormal Krylov basis $\{|O_n\rangle\}_{n\geq 0}$ via the three-term recursion $b_{n+1}\,|O_{n+1}\rangle = (-iK_{\mathcal{N}})\,|O_n\rangle - a_n|O_n\rangle - b_n|O_{n-1}\rangle$ with Lanczos coefficients $\{b_n\}_{n\geq 1}$. Define the *Krylov spread complexity*

$$C_k(s) \;:=\; \sum_{n\geq 0} n\, \big|\langle O_n \,|\, e^{-iK_{\mathcal{N}} s} \,|\, \mathrm{seed}\rangle\big|^2. \tag{3}$$

Reductions. (i) On a finite-trace corner $p\mathcal{N}p$ with $\mathrm{tr}_{\mathcal{N}}(p) < \infty$, Definition 2 reduces to the finite-dimensional Krylov complexity of Parker et al. [6]. (ii) Restriction of the seed to $\mathcal{A}(D_O)_{\mathrm{FRW}} \otimes \mathbf{1}$ ("ungauged" limit, dropping the $\mathbb{R}$-factor) recovers the type III_1 modular Krylov complexity of [2]. (iii) On a type II_1 algebra (e.g. the de Sitter static-patch construction of [4]) the same definition reproduces the DSSYK / boundary-Krylov construction of Aguilar-Gutiérrez [1] after substituting the type II_1 trace.

3.2 Status

Remark 3. Definition 2 is well-posed under standard hypotheses on $\mathcal{N}$ and on the smooth seed (the seed lies in the analytic vectors of $K_{\mathcal{N}}$; the Krylov basis is orthonormal in $L^2(\mathcal{N}, \mathrm{tr}_{\mathcal{N}})$; the spread complexity is a normalised moment of the Lanczos return amplitude as in Vardian [2] and [1]). What we do *not* prove here is the universal operator-growth hypothesis (UOGH) on type II_∞ KMS states,

which would underwrite a $b_n = O(n)$ scaling and the $e^{2\pi s}$ asymptotic of $C_k(s)$ on $\mathcal{N}$ analogous to Parker et al. [6] and CMPT [2]. The asymptotic slope $\lambda_L^{\mathrm{mod}} = 2\pi$ is taken as input from the type III_1 subalgebra: in the conformal-pullback setting of Theorem 1, the seed and the Lanczos sequence in $L^2(\mathcal{N}, \mathrm{tr}_{\mathcal{N}})$ project onto the type III_1 Lanczos sequence of [2] on $L^2(\mathcal{A}(D_O)_{\mathrm{FRW}}, \Omega_{\mathrm{FRW}})$ via the conditional expectation $E : \mathcal{N} \to \mathcal{A}(D_O)_{\mathrm{FRW}}$, which preserves the late-modular-time growth rate. A direct II_∞-intrinsic proof of $\lambda_L^{\mathrm{mod}} = 2\pi$ is open (see Section 7).

4 Main theorem: Krylov–Diameter Correspondence

The Bisognano–Wichmann/Hislop–Longo modular Hamiltonian on a spacelike-symmetric Minkowski diamond of proper radius R_{proper} has, at the centre $\mathbf{x}_c$, a local Unruh temperature

$$T_{\mathrm{loc}}(\mathbf{x}_c) \;=\; \frac{1}{2\pi\, R_{\mathrm{proper}}}, \qquad \text{equivalently} \qquad \left.\frac{\mathrm{d}s}{\mathrm{d}\tau_{\mathrm{proper}}}\right|_{\mathbf{x}_c} \;=\; \frac{1}{2\pi\, R_{\mathrm{proper}}}, \tag{4}$$

where s is the Tomita–Takesaki modular parameter (the parameter of $\Delta_{\mathcal{N}}^{is}$) and τ_{proper} is proper time of an observer at the centre. This is the Casini–Huerta–Myers relation [3]; here $R_{\mathrm{proper}} = R_{\mathrm{proper}}(\eta_c)$ as defined by (2), since the central Cauchy slice intersects the centre worldline at proper time corresponding to η_c.

Theorem 4 (Krylov–Diameter Correspondence on the type II_∞ FRW crossed product)**.** *Let $a(\eta)$ be a smooth strictly positive scale factor on $\eta \in I$, D_O a doubly-bounded comoving causal diamond with $\overline{D_O} \subset I$, $\mathcal{N} = \mathcal{A}(D_O)_{\mathrm{FRW}} \rtimes_\sigma \mathbb{R}$ the type II_∞ factor of Theorem 1, C_k the Block A modular Krylov spread complexity of Definition 2 on a smooth seed $|\mathrm{seed}\rangle \in L^2(\mathcal{N}, \mathrm{tr}_{\mathcal{N}})$, and t the comoving observer's cosmic time at the centre of D_O. Then, in the late-modular-time asymptotic regime under the working assumption that the universal slope $\lambda_L^{\mathrm{mod}} = 2\pi$ of CMPT [2] carries over to $\mathcal{N}$ via Remark 3,*

$$\boxed{\frac{1}{C_k(t)}\,\frac{\mathrm{d}C_k(t)}{\mathrm{d}t} \;=\; \frac{1}{R_{\mathrm{proper}}(\eta_c)} \;+\; O\big(C_k^{-1}\big),} \tag{5}$$

where $R_{\mathrm{proper}}(\eta_c) = a(\eta_c)\, R_{\mathrm{conf}}$ is the proper diameter radius at the central Cauchy slice. Both sides of (5) are spacetime scalars on the central worldline; the identity is manifestly invariant under reparametrisations $t \mapsto t'(t)$ of the worldline parameter.

Proof sketch. By [2, Eq. (4.16)/(5.4)] and Remark 3, the Lanczos coefficients on the type III_1 sub-flow satisfy $b_n \sim \pi n$ in modular-time units, hence $C_k(s) \sim e^{2\pi s}$ at late modular time and $(1/C_k)\,\mathrm{d}C_k/\mathrm{d}s = \lambda_L^{\mathrm{mod}} = 2\pi$. By (4), on the central worldline, $\mathrm{d}s/\mathrm{d}t = 1/(2\pi\, R_{\mathrm{proper}}(\eta_c))$ where $t = \tau_{\mathrm{proper}}$ for a comoving observer in FRW. The chain rule then gives $\frac{1}{C_k}\,\frac{\mathrm{d}C_k}{\mathrm{d}t} = 2\pi \cdot \frac{1}{2\pi R_{\mathrm{proper}}(\eta_c)} = \frac{1}{R_{\mathrm{proper}}(\eta_c)}$, up to corrections of order C_k^{-1} from the subleading late-modular-time expansion of the spread complexity. □

Remark 5 (Gauge invariance)**.** The identity (5) is invariant under reparametrisation of the worldline parameter (e.g. $t \to \eta$ or any other smooth monotone reparametrisation): both $(1/C_k)\,\mathrm{d}C_k/\mathrm{d}t$ and $1/R_{\mathrm{proper}}(\eta_c)$ transform identically by the chain rule, and the second is a pointwise spacetime scalar. The traversal through cosmic time vs. conformal time vs. modular time is purely a matter of which parameter one differentiates with respect to; the underlying geometric–algebraic equality is the same. We verified this symbolically (sympy, script `piste1_gauge_audit.py`); see [7] for details.

Remark 6 (What this is *not*)**.** (5) is *not* a universal Hubble–Krylov identity. The Hubble parameter $H(t) := (1/a)\,\mathrm{d}a/\mathrm{d}t\big|_{\text{centralworldline}}$ agrees with the right-hand side of (5) only in two of the four era/diamond combinations tabulated in Section 5, and *disagrees by a factor of* 1/2 for matter + past-light-cone diamond. The clean statement is geometric (proper diameter radius), not Friedmann (Hubble expansion rate); see also the gauge audit cited in the title footnote. Conversely, as a consistency check, the special cases for which $H(t) = 1/R_{\text{proper}}(\eta_c)$ does hold (radiation + past-light-cone, de Sitter + Hubble-horizon) are recovered as Friedmann arithmetic, not as deeper structural features of $\mathcal{N}$.

5 Era-by-era special cases

We tabulate the right-hand side of (5) for the three backgrounds covered by [7, Thms. 3.5, 3.6, Cor. 3.7], combined with two natural conventions for the diamond: (**PLC**) the past-light-cone diamond $\eta_i \to 0^+$, $\eta_f = 2\eta_{\text{obs}}$, $R_{\text{conf}} = \eta_{\text{obs}}$ (only meaningful when the lower limit is regular; for de Sitter we use the analogous past-cone with $\eta_i \to -\infty$); (**H**) the Hubble-horizon diamond defined by $R_{\text{proper}} = 1/H(t)$ at the central slice. Both ratios in the final column are in cosmic time t; the entries are sympy-verified (script `piste1_gauge_audit.py`, see [7]).

Era	$a(\eta)$	$H(t)$	Diamond	$R_{\text{proper}}(\eta_c)^{-1}/H(t)$
Radiation	$a_0\,\eta$	$1/(2t)$	PLC	1 (exact)
Matter	$a_0\,\eta^2$	$2/(3t)$	PLC	1/2
Matter	$a_0\,\eta^2$	$2/(3t)$	H	1 (tautological)
de Sitter	$-1/(H_{\text{dS}}\,\eta)$	H_{dS}	H	1 (tautological)

The reading of the table is as follows. The Hubble identity $H(t) = (1/C_k)\,\mathrm{d}C_k/\mathrm{d}t$ holds: (i) exactly, by Friedmann arithmetic, for radiation + PLC, since $R_{\text{proper}}(\eta_c) = a(\eta_c)\,\eta_{\text{obs}} = a_0\eta_{\text{obs}}^2 = 2t$ in radiation FRW; (ii) tautologically for matter + Hubble-horizon and for de Sitter + Hubble-horizon, where the diamond is *defined* by $R_{\text{proper}} = 1/H$; and *fails by a factor of* 2 for matter + PLC, where $R_{\text{proper}}(\eta_c) = a(\eta_c)\,\eta_{\text{obs}} = a_0\eta_{\text{obs}}^3 = 3t$ gives $1/R_{\text{proper}}(\eta_c) = 1/(3t)$ vs. $H(t) = 2/(3t)$.

Remark 7 (On the matter + PLC factor of 1/2)**.** The discrepancy in the matter + PLC row is a feature of the diamond convention, not of the algebra. Different conventions for R_{conf} as a function of η_i, η_f give different values of R_{proper}, hence of the right-hand side of (5). The *rate* on the algebraic side (the left-hand side) is unambiguously $1/R_{\text{proper}}(\eta_c)$ for the chosen diamond; the *rate* on the geometric side, $H(t)$, is intrinsic to the background but not to the diamond. There is no contradiction; the identity (5) is not the equation $H(t) =$ Krylov rate, which is the era-and-diamond-conditional corollary discussed above.

6 Comparison with CMPT and the modular Lyapunov 2π

CMPT [2] establish, on the type III_1 algebra of a two-dimensional CFT in a vacuum-like state, the universal late-modular-time slope

$$\frac{1}{C_k(s)}\,\frac{\mathrm{d}C_k(s)}{\mathrm{d}s} \;=\; \lambda_L^{\text{mod}} \;=\; 2\pi, \tag{6}$$

where s is the Tomita–Takesaki modular parameter intrinsic to the algebra (the parameter of Δ^{is}). The constant 2π is the modular analogue of the universal upper bound on operator growth [6] and is independent of any background-spacetime time coordinate.

The present Theorem 4 converts (6) into a statement about cosmic time t along the comoving worldline of the diamond's centre. The conversion is the Casini–Huerta–Myers relation (4) between modular and proper time at the centre of a spacelike-symmetric diamond. The factor 2π in (6) cancels the factor 2π in (4), leaving the clean geometric statement (5). The Krylov complexity in (5) is on the type II_∞ crossed product $\mathcal{N}$, not on the type III_1 subalgebra $\mathcal{A}(D_O)_{\mathrm{FRW}}$; the Block A definition of Section 3 ensures that the late-modular-time slope is inherited.

The closest published precedent for a Krylov construction on a type II algebra with cosmological motivation is Aguilar-Gutiérrez [1], where the type II_1 DSSYK Krylov complexity is computed on the boundary Hartle–Hawking seed. Theorem 4 is the type II_∞ extension of that programme adapted to the FRW geometric setting and specialised to the spread complexity at the diamond centre.

7 Open problems

Open Question 8 (II_∞-intrinsic proof of $\lambda_L^{\mathrm{mod}} = 2\pi$)**.** Prove the universal operator-growth hypothesis on type II_∞ KMS states with respect to the canonical trace $\mathrm{tr}_{\mathcal{N}}$, in particular establishing $b_n = O(n)$ on $L^2(\mathcal{N}, \mathrm{tr}_{\mathcal{N}})$ for the lifted modular Hamiltonian $K_{\mathcal{N}}$. This would close the gap noted in Remark 3: the slope $\lambda_L^{\mathrm{mod}} = 2\pi$ in Theorem 4 is currently inherited from the type III_1 subalgebra via a conditional-expectation argument; a direct II_∞ proof would also clarify whether the constant 2π is robust under the addition of a non-trivial cocycle (e.g. a small mass perturbation).

Open Question 9 (Diamond-convention-independent geometric invariants)**.** The right-hand side of (5) depends on $R_{\mathrm{proper}}(\eta_c)$, hence on the choice of diamond. Are there *diamond-convention-independent* combinations of C_k and its derivatives that recover purely intrinsic FRW invariants (e.g. $H(t)$, $\dot{H}$, $\dot{H}/H^2$) as geometric–algebraic identities? A natural candidate would involve the running of R_{proper} across a one-parameter family of nested diamonds.

Open Question 10 (Big Bang regime, $\eta_i \to 0^+$)**.** The companion note [7] (open-questions section, "Big Bang – algebraic structure at $\eta_i = 0$") flags three obstructions to the conformal-pullback construction in 4D as $\eta_i \to 0^+$: (a) failure of the conformal-rescaling bijection $f \mapsto a^3 f$ at the Big Bang, (b) IR singularity of the conformal-vacuum two-point function, (c) non-Hadamard boundary behaviour. In 2D all three vanish (as verified by the `eta_zero_2d_opus.py` script of [7]); in 4D none are addressed by the present argument. The PLC diamond of the radiation row in Section 5 is a borderline case which heuristically extends to $\eta_i \to 0^+$ but which the present proof does not establish.

Open Question 11 (Beyond the leading order, and non-PLC diamonds)**.** $1/R_{\mathrm{proper}}(\eta_c)$ in (5) is the leading late-modular-time slope; the $O(C_k^{-1})$ correction is seed-dependent. An all-orders expansion of $C_k(t)$ in the cosmic time variable, in terms of geometric invariants of the diamond and its central worldline, would give a genuine cosmological observable. A related question: for an arbitrary spacelike-symmetric diamond inside D_O, under what intrinsic conditions on the diamond does the ratio $1/(R_{\mathrm{proper}}(\eta_c)\, H(t))$ reduce to a clean closed form (e.g. a function of the equation-of-state parameter w alone)?

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large-latest, 2026-05-02) followed by a sympy-verified gauge audit; the technical content does not depend on either of these, but the *statement* does. We thank the authors of [1, 2] for opening the modular-Krylov programme on which this note depends, and the authors of [4, 8] for the type II_∞ crossed-product machinery without which Theorem 1 (and hence Theorem 4) would not be available.

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